Dimension Pool: Deadlock-Safe Buffer Pooling Across Opposing Inports on a 3D Torus

Lab 4 Project Report — Topic 2: Flow Control

August 28, 2026

Abstract

Classical virtual-channel (VC) allocation statically splits each router's buffers per inport, so when traffic through a dimension is momentarily one-sided — the common case on a torus — the cold direction's VCs sit idle while the hot direction head-blocks. This paper presents *Dimension Pool* (DP), a flow-control mechanism in which the two opposing inports of each dimension ($E{+}W$, $N{+}S$, $U{+}D$) share their pooled VCs under a joint occupancy cap S, letting a hot direction borrow budget the cold direction is not using — without relocating any packet and without touching the deadlock-freedom substrate. DP composes with two independently deadlock-free baselines on the same 4×4×4 Garnet Torus3D substrate: dimension-order routing protected by the Critical Bubble Scheme (DP-CBS) and minimal adaptive routing with escape VCs (DP-ESC). Under an equal-usable-storage methodology (a DP configuration is compared against a plain baseline with the same usable slots per dimension pair), DP-ESC gains **+21–31%** peak stable throughput on all five synthetic patterns at budget $C{=}6$, and DP-CBS gains **+20.9%** on xopposite at $C{=}6$. All 1006 runs, including six 200 000-cycle saturation probes, are deadlock-free. We also report the mechanism's costs honestly: the shared cap loses up to −32.6% at equal VC count under two-sided load, DP-CBS turns slightly negative (−2.7%) once buffers are plentiful ($C{=}8$), and gains on symmetric uniform-random traffic are near zero — pooling pays exactly where demand between directions is imbalanced.

1 Introduction

On a torus, the $+d$ and $-d$ directions of a dimension are rarely equally loaded at a given router and moment: adversarial patterns (xopposite, tornado) are strictly one-sided per ring, and even balanced patterns are one-sided *locally* over short windows. Yet the standard Garnet buffer organization dedicates every VC to one inport, i.e. to one direction. The result is a structural inefficiency: the hot side saturates on its private slots while the opposite side's slots are provably idle.

Dimension Pool (DP) removes this split at the accounting level. The pooled VCs of the two opposing inports form one budget, admission into any pooled VC requires the *joint* occupancy to stay below a cap S, and nothing else changes: no packet ever moves between buffers, the datapath and crossbar are untouched, and deadlock freedom is delegated to a substrate that is never pooled. Highlights:

- **A new flow-control primitive.** DP is neither a VC reassignment scheme nor a shared physical buffer (DAMQ): it is a counter-only occupancy budget spanning two inports, cheap enough to sit in the allocator's existing credit checks.
- **Two deadlock-safety compositions.** DP-CBS runs the Critical Bubble Scheme [4] verbatim on a private dedicated tranche (a "depth-domain" instance of Duato's argument [3]); DP-ESC exempts escape VCs [1, 3] from the pool so Duato's theorem applies unchanged. In both, the pool is a pure performance resource: pooled admission may fail at any time without risking deadlock.
- **An equal-usable-storage methodology.** Every DP configuration is compared against a plain baseline with the same usable slots per dimension pair, so measured wins come from pooling flexibility, not extra buffers. Claims are further stress-tested by cap-falsification reruns ($S{=}1$) and a bit-identical baseline-purity check.
- **Honest accounting of costs.** We report where DP loses (equal-VC two-sided load, buffer-rich regimes) and where it is neutral (uniform random), alongside the +21–31% headline gains.

The rest of this paper is organized as follows. Section 2 reviews the platform and the two deadlock-avoidance disciplines DP builds on — the Critical Bubble Scheme and escape-VC adaptive routing — which double as the evaluation baselines. Section 3 presents the design and its safety argument, Section 4 the implementation, Section 5 the experiments, and Section 6 the analysis of why and when pooling pays.

2 Background

2.1 Platform

All work targets gem5 Garnet (standalone), 4×4×4 Torus3D (64 nodes), synthetic traffic on control vnet 0 with single-flit packets. Control VCs hold one packet each (`buffers_per_ctrl_vc` = 1), so a buffer *slot*, a VC, and a packet-in-flight coincide, and all flow-control bookkeeping in this paper is pure counting. The harness injects at most 0.5 packets/node/network-cycle, which acts as an offered-load ceiling in the experiments.

2.2 Deadlock avoidance on torus rings

The wrap-around links that give a torus its low diameter also close a cyclic channel dependency on every ring, so dimension-order routing (DOR), safe on a mesh, deadlocks on a torus [5]. Two classical disciplines restore safety; both are implemented and validated on this platform, and they double as the two baseline families of the evaluation (§5).

Critical Bubble Scheme (family A). Bubble flow control [5] keeps DOR and instead constrains buffer occupancy: as long as every directed ring retains one free slot (a *bubble*), some packet on the ring can always advance. The Critical Bubble Scheme (CBS) [4] implements this at one-slot cost: a single slot per directed ring is marked *critical* and may only be consumed by in-transit packets, never by injection, so the ring can always drain. Our CBS implementation is itself a flow-control result: on transpose it lifts peak stable throughput from 0.224 to 0.258 (+15%) over unprotected DOR — the injection-side entry rule throttles the sources just before saturation trees form — and it is zero-overhead on the other patterns, where its curves coincide with plain DOR. Family A baselines (A1, A3 in Table 2) run DOR + CBS with 3 and 4 VCs per inport.

Minimal adaptive routing + escape VCs (family B). The alternative discipline gives up deterministic routing: minimal-quadrant adaptive routing over the torus wrap links, with the last VC of each inport reserved as an escape channel routed by deadlock-free (acyclic-CDG) Mesh3D XYZ routing. By Duato's theorem [1, 3], a packet blocked on adaptive VCs can always eventually take the escape path, so the adaptive tranche may be arbitrarily congested. On this platform the adaptive router reaches the 0.5 offered-load ceiling on all five synthetic patterns. Baselines B1, B3, B5 run this design with 2, 3, 4 VCs per inport.

Both disciplines, like classical VC flow control itself [2], leave one structural decision untouched: every buffer slot is statically owned by one inport, i.e. by one direction of one dimension. That static split is the inefficiency DP removes.

3 Design: the Dimension Pool

DP's entire mechanism is one counter per dimension pair and one admission inequality — no new arbiter, no table, no pipeline stage, no datapath change. This minimality is not an accident of scope but the central design property: every alternative that buys the same flexibility (DAMQ-style shared physical buffers, dynamic VC reassignment) does so with multi-ported storage or packet relocation, and those are precisely the structures a deadlock argument must then re-examine. Keeping the mechanism to a counter is what keeps the safety argument (§3.3) a clean composition with an untouched substrate, and what makes the hardware cost (§3.4) a sideband wire rather than an SRAM redesign. This section develops the design in full: the pool structure (§3.1), the admission and retry semantics with the placement of the check (§3.2), the deadlock-safety composition (§3.3), and the hardware cost (§3.4).

3.1 Pool, cap, and tranches

For each router, dimension d, and control vnet, define

$$pool\_occ(R, d, v) \;=\; occ(\text{inport } +d) + occ(\text{inport } -d), \tag{1}$$

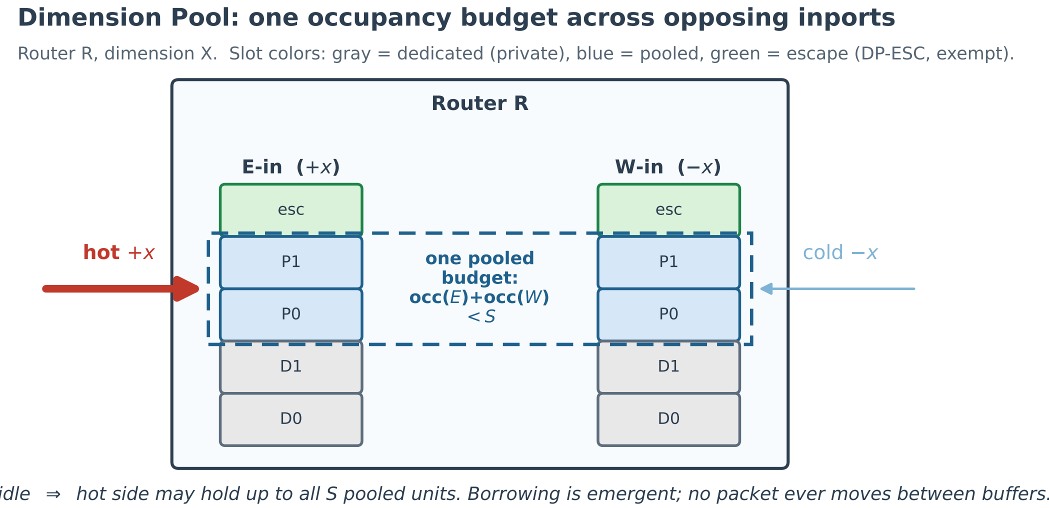

Figure 1: The Dimension Pool at one router, dimension X. Each inport keeps its private per-VC FIFOs; only the *accounting* of the blue pooled slots is joined under the single budget $occ(E) + occ(W) < S$. Gray dedicated slots (DP-CBS) and the green escape VC (DP-ESC) are outside the pool.

where occ counts packets currently holding a *pooled* VC at that inport, from allocation grant until the freeing credit returns. Admission into any pooled VC requires

$$pool\_occ < S \qquad (S = \texttt{-dp-shared-cap}). \tag{2}$$

There is no "enable event" and no slot transfer: borrowing is emergent. When one side is idle it contributes ≈ 0 to $pool\_occ$, so the hot side can occupy up to all S units of the pair budget; under two-sided load the two inports compete for the same budget.

Figure 1 illustrates the structure, and a worked example shows the borrowing. Under xopposite traffic every packet at this router crosses in $+x$, so the $-x$ inport holds no pooled slot and $pool\_occ$ reduces to the hot inport's own occupancy. The hot inport may therefore hold its dedicated tranche *plus the entire pair budget* — at r=2, S=2 that is $2 + 2 = 4$ usable lanes, where the static equal-storage design (C=6, i.e. 3 VCs per inport) caps it at 3. Nothing was signalled or moved to enable this; and when the pattern reverses, the hot side's occupancy drains within a credit round trip and the same budget is simply available to the other side.

VCs are partitioned into tranches that are *ID-pegged* (fixed at configuration time, a property of the VC id — count-based accounting would break the CBS bubble invariant, which needs the identity of the critical slot to be stable):

- **DP-CBS** (on family A): VC ids $[0, r)$ are a *dedicated* tranche private to the inport (r=2 throughout); ids $[r, V)$ are pooled.
- **DP-ESC** (on family B): the adaptive window $[0, V - 1)$ is pooled; the escape VC is *exempt* — never counted in $pool\_occ$, never denied by the cap.

When both a pooled and a dedicated VC are free, the allocator prefers the pooled one: spending shared budget before private reserve maximizes bubble mobility in the dedicated sub-ring. (The design mirrors pooled liquidity with a private per-direction reserve in payment-channel networks; the reserve guarantees progress, the pool provides the flexibility.)

3.2 Admission and retry semantics

The pool check is a *pre-flight* credit-style test: a packet that fails admission never leaves its upstream VC; there is no NACK, drop, or retransmission. Switch allocation re-arbitrates every cycle, so a denied packet simply retries. The two variants differ in where the primary check sits (Figure 2):

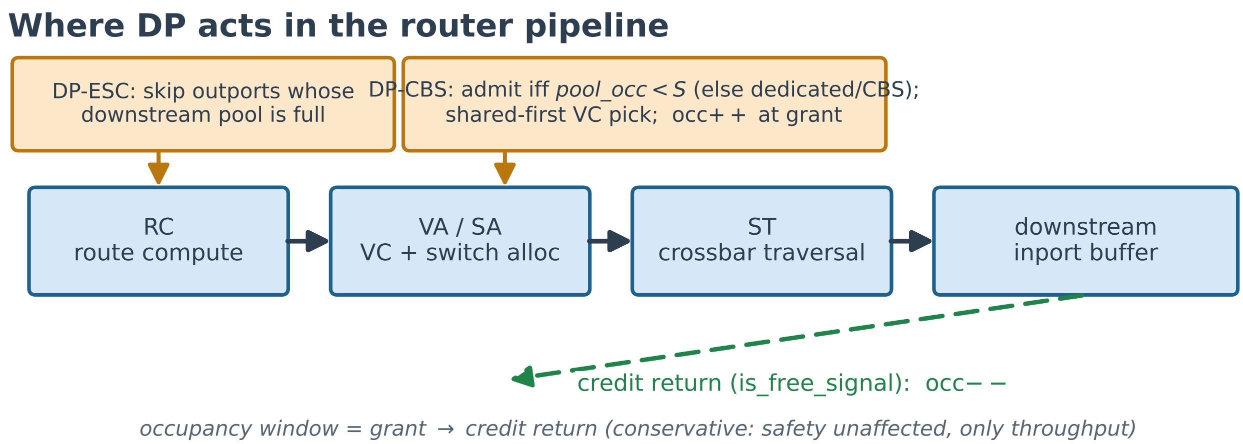

Figure 2: Where DP acts in the router pipeline. DP-ESC filters route candidates at route computation (with a belt check at switch allocation for the intervening race); DP-CBS checks admission in the switch allocator. The occupancy window spans allocation grant to credit return.

- **DP-ESC**: the primary check is a candidate filter inside the adaptive route computation — an outport whose downstream pool is full is skipped exactly like an outport with zero free credits, so the packet either picks another minimal adaptive direction or falls through to the escape VC via the existing machinery. A belt check in `send_allowed` covers the race where the pool fills between route computation and switch allocation. This placement preserves Duato's premise: a packet waiting on adaptive resources always still has the escape path available.
- **DP-CBS**: DOR fixes the outport, so the check sits in the switch allocator's admission rule; on failure the packet waits for, or falls back to, the dedicated tranche under normal CBS entry rules.

3.3 Deadlock safety

DP only ever *restricts* admission to pooled slots and changes no routes, but pooling alone is not safe. Suppose one kept only the cap rule with no discipline inside the private slots: on a single ring under DOR, every node injecting in one direction can fill *all* private and pooled slots with packets each waiting for the next hop — the cap is respected at every router, yet the ring holds no free slot and deadlocks. A cap bounds *how many* slots each side holds; deadlock freedom needs an invariant about *which* slot stays free. DP therefore composes with a substrate that provides exactly that invariant:

DP-CBS: depth-domain Duato. The dedicated window $[0, r)$ runs CBS verbatim as a private sub-ring of depth r: ring entry toward a marked inport requires two free dedicated slots, and in-ring transit may fill the last free dedicated slot only if the mover itself resides in a dedicated VC — otherwise a packet sitting in a pooled VC could consume the bubble while returning its own slot to the pool, where the opposite direction may claim it, and the sub-ring would lose its bubble permanently. The pooled window is the "adaptive layer" of a Duato construction, transplanted from VC class to buffer depth: it may be arbitrarily congested because every packet in it is also eligible for the dedicated (escape) layer at every hop.

DP-ESC: Duato verbatim. Escape VCs are exempt from the pool, so the escape network's acyclic channel-dependency graph is untouched and Duato's theorem applies unchanged.

Conservative accounting. *pool_occ* is incremented at switch-allocation grant (a reservation, before the flit moves) and decremented when the freeing credit returns, so occupancy spans grant $\rightarrow$ credit-return and slightly overestimates instantaneous buffer occupancy. Because the pool is never in the safety path, this staleness can only cost throughput, never correctness. Same-cycle over-admission is excluded by gem5's sequential router wakeup together with a configuration guard requiring all torus dimensions ≥ 3 (a size-2 dimension would let both outports of one router feed the same downstream pool in one cycle).

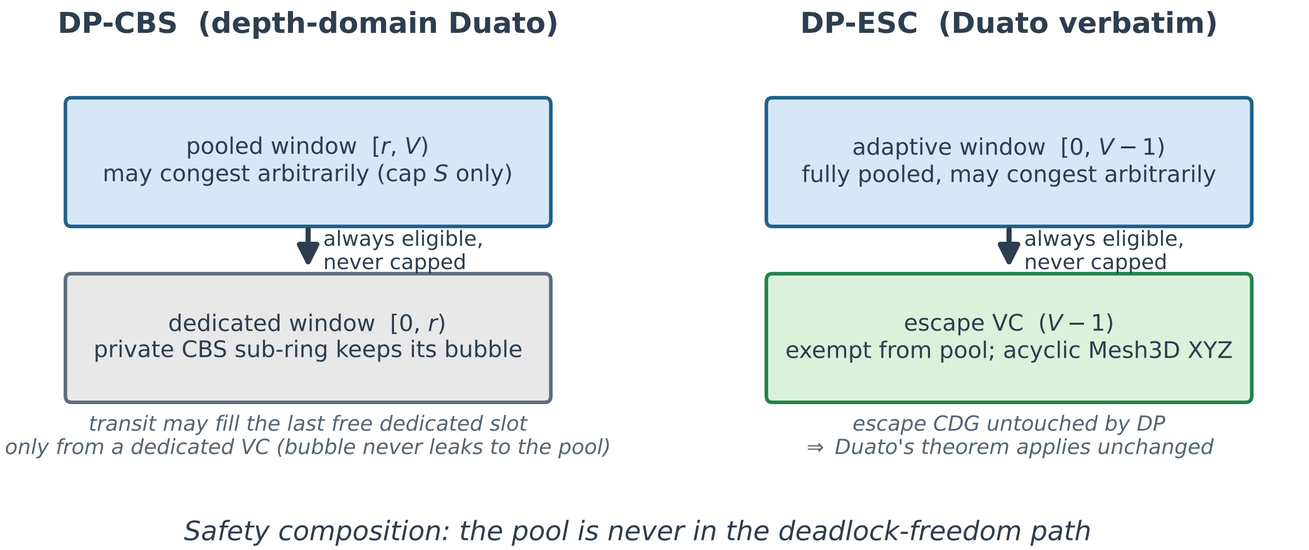

Figure 3: The two safety compositions. In both, the pooled tranche may congest arbitrarily because every packet in it remains eligible at every hop for an unpooled resource that keeps its own invariant: the private CBS sub-ring (left — Duato's construction transplanted from VC class to buffer depth) or the exempt escape VC (right — Duato's theorem verbatim).

3.4 Hardware cost

DP's state is three $\lceil \log_2(S+1) \rceil$-bit counters per router per control vnet (one per dimension pair) plus an adder and comparator in the allocator — negligible next to the buffers themselves. The real new item is *wiring*: the joint occupancy of a downstream router's two inports must be visible to two different upstream routers. In simulation the registry is a zero-latency oracle; a hardware realization keeps the counter at the downstream router and distributes a token-conserved pool-credit sideband of roughly one wire per link per pooled vnet, piggybacking on the existing credit channel. Because tranches are ID-pegged, an implementation over-provisions VC ids (build 2× pooled, use $\leq S$) rather than porting a dual-ported shared SRAM — cheap for one-flit control slots. Datapath, crossbar, and pipeline are untouched.

4 Implementation

DP is implemented in gem5 v23.0 Garnet behind `-enable-dp` (default off; all baseline paths bit-identical when disabled, verified in §5.3).

File	Change
`GarnetNetwork.py` / `Network.py`	`enable_dp`, `dp_reserve` (r), `dp_shared_cap` (S) parameters and CLI flags
`GarnetNetwork.hh/.cc`	occupancy registry `m_dp_shared_occ[router][dir][vnet]`, helpers, stats, config validation
`SwitchAllocator.cc`	DP-CBS admission rule, shared-first VC selection, grant-time increment, belt check
`OutputUnit.cc`	decrement on the returning `is_free_signal` credit
`RoutingUnit.cc`	pool-full candidate filter in the adaptive route computation (DP-ESC)

Table 1: Implementation map. Both bookkeeping sites address the downstream buffer owner symmetrically.

New statistics: `dp_shared_grants` (allocations landing in pooled VCs) and `dp_pool_blocks` (admissions denied by the cap). Configuration validation rejects unsupported operating points with `fatal`: torus dimensions ≥ 3, control vnets only, routing algorithms 3/4 only, no wormhole, $S \in [1, 2 \times \text{pooled per inport}]$.

5 Experiments

5.1 Methodology: equal usable storage

A DP configuration with d dedicated VCs per inport and cap S exposes at most $C = 2d + S$ usable slots per dimension pair. Each DP mode is compared against the plain baseline whose total storage per pair equals that C (Table 2); DP carries more physical VC ids, but their concentration on the hot inport is exactly the mechanism under test, so wins cannot come from extra buffers.

Family	Baseline	DP configuration	DP parameters	C
A (DOR + CBS)	A1: vcs=3	A2: vcs=4	r=2, S=2	6
	A3: vcs=4	A4: vcs=6	r=2, S=4	8
B (adaptive + escape)	B1: vcs=2	B2: vcs=3	S=2	4
	B3: vcs=3	B4: vcs=5	S=4	6
	B5: vcs=4	B6: vcs=7	S=6	8

Table 2: Equal-usable-storage pairs. $C = 2 \cdot \text{dedicated} + S$ usable slots per dimension pair. The B C=4 and B C=8 pairs are swept but excluded from headline gains — §5.2 explains why they are uninformative; they return as attribution and regime-boundary evidence (§5.3, §6).

Sweep: five patterns (xopposite, tornado, transpose, neighbor, uniform random) × 20 injection rates (0.05–1.00) × 10 modes = 1000 runs of 10 000 cycles. Throughput is received packets/node/cycle; a point is *stable* when ≥ 90% of offered load is accepted; the offered-load ceiling of the harness is 0.5 packets/node/cycle.

5.2 Results

Pair	xopposite	tornado	transpose	neighbor	uniform
A, C=6	**+20.9%**	ceil	+1.4%	ceil	≥+2.6%
A, C=8	−2.7%	ceil	+1.3%	ceil	ceil
B, C=6	**+21.3%**	≥+31.0%	≥+29.2%	≥+24.9%	≥+28.5%

Table 3: Peak stable throughput gain of DP over its equal-storage baseline, informative pairs only. "ceil": both sides saturate the 0.5 injection ceiling (no information). "≥": the DP side is at the ceiling, so the gain is a lower bound.

Two sweep rows are omitted from Table 3 as uninformative rather than unfavorable. At B C=4 the baseline B1 is credit-round-trip starved: with a single usable adaptive lane per inport and a ∼5-cycle VC turnaround it saturates near 0.195 on *all five* patterns, so the raw +95–103% of B2 over it measures baseline starvation, not pooling. (The honest content of that row is budget efficiency: on one-sided patterns B2 is *bit-identical* to the C=6 baseline B3 at every rate — a 4-slot pair budget doing the work of a 6-slot static design — and B2 returns as attribution evidence in §5.3.) At B C=8 every configuration saturates the harness's 0.5 offered-load ceiling on every pattern, so no relative statement survives.

Figure 4 shows the two headline latency–throughput cells and Figure 5 draws the gains. Three readings, in decreasing order of strength:

1. **DP-ESC at C=6: +21–31% on every pattern** (mostly lower bounds), including uniform random and including the one cell (xopposite) with no ceiling excuse. This is the headline claim.
2. **DP-CBS: one clean win, honestly bounded.** +20.9% on xopposite at C=6; ≥+2.6% at zero cost on uniform; and −2.7% at C=8 on xopposite — once buffers are plentiful, the fixed depth-2 safety sub-ring stops paying for its rigidity. DP-CBS is a scarce-buffer technique.
3. **The cap has a price under two-sided load.** At equal VC count (B2 vs. B3, same 3 VCs), the shared budget costs up to −32.6% on xopposite: both directions compete for one budget, and a hot side can deny the other side admission even while physically free pooled VCs exist on the denied side. Equal-storage flexibility is bought with cap contention.

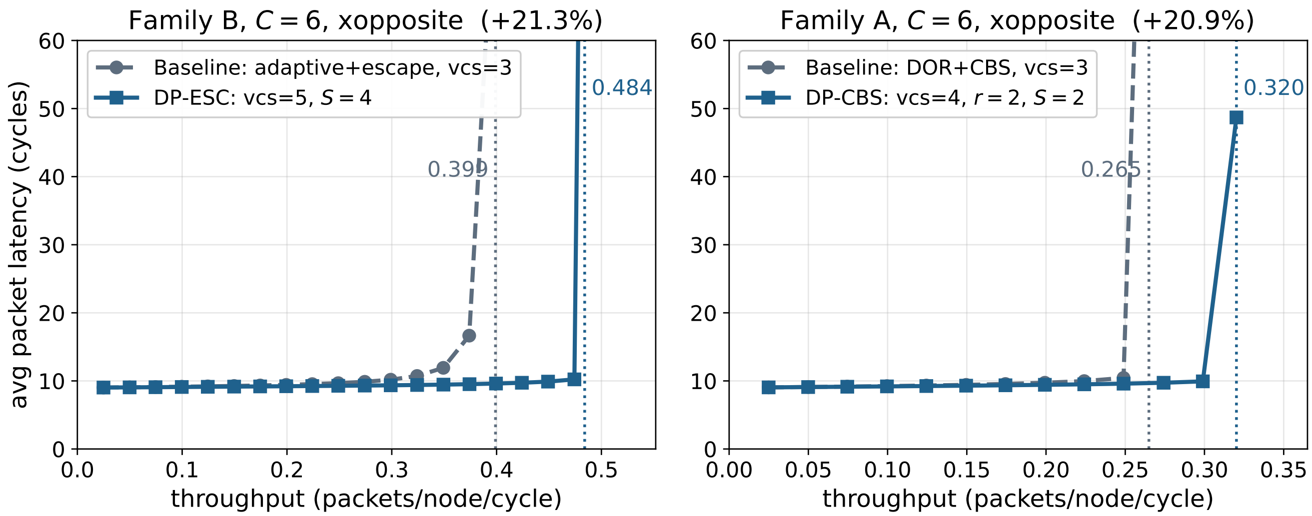

Figure 4: Latency–throughput curves for the two headline cells (xopposite, C=6; stable points only, dotted verticals mark peak stable throughput). Left: DP-ESC, +21.3%. Right: DP-CBS, +20.9%. In both, DP holds flat latency past the baseline's collapse point at equal usable storage.

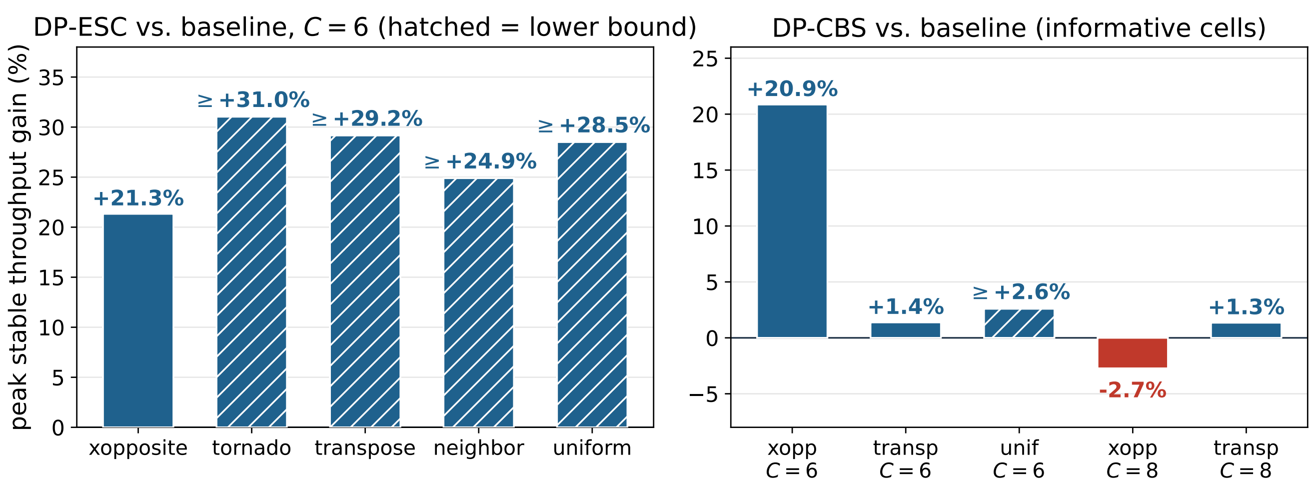

Figure 5: Table 3, drawn. Left: DP-ESC at C=6 across all five patterns; hatched bars are lower bounds (the DP side is at the 0.5 ceiling). Right: the informative DP-CBS cells, including the honest negative at C=8.

5.3 Validation and attribution

Deadlock freedom. Zero deadlocks in all 1006 runs, including six 200 000-cycle saturation probes at injection rate 1.0 on the most at-risk DP configurations (A2, A4, B2 on transpose and xopposite) — the empirical leg of §3.3.

Baseline purity. The post-DP binary reproduces the pre-DP sweep's baseline results *bit-identically* at all 160 shared rate points, so no measured gain comes from accidental changes to baseline code paths.

Cap falsification. If the gain comes from the pooled budget, halving the cap (S=1) should destroy it:

Case	Baseline	DP S=2	DP S=1	Reading
A2, xopposite	0.265	0.320	0.251	collapses → gain attributable to the pool budget
B2, tornado	0.195	0.381	0.297	∼45% of the gain vanishes → budget is a partial contributor
B2, neighbor	0.197	0.400	0.400	unchanged — but via 128k escape transitions (0 at S=2)

Table 4: S=1 falsification (peak stable throughput). The neighbor row gives the honest attribution for one-sided B-family traffic: a hot-inport *lane-count* effect provided jointly by the pool budget and the escape exemption.

Instrumentation caveat. For DP-ESC, pool pressure is absorbed by the routing-time candidate filter, which has no counter, so `dp_pool_blocks` reads zero in family B by construction; the pressure is visible as `escape_transitions` and a changed direction mix instead (Figure 6 shows the pool activity counters that are populated).

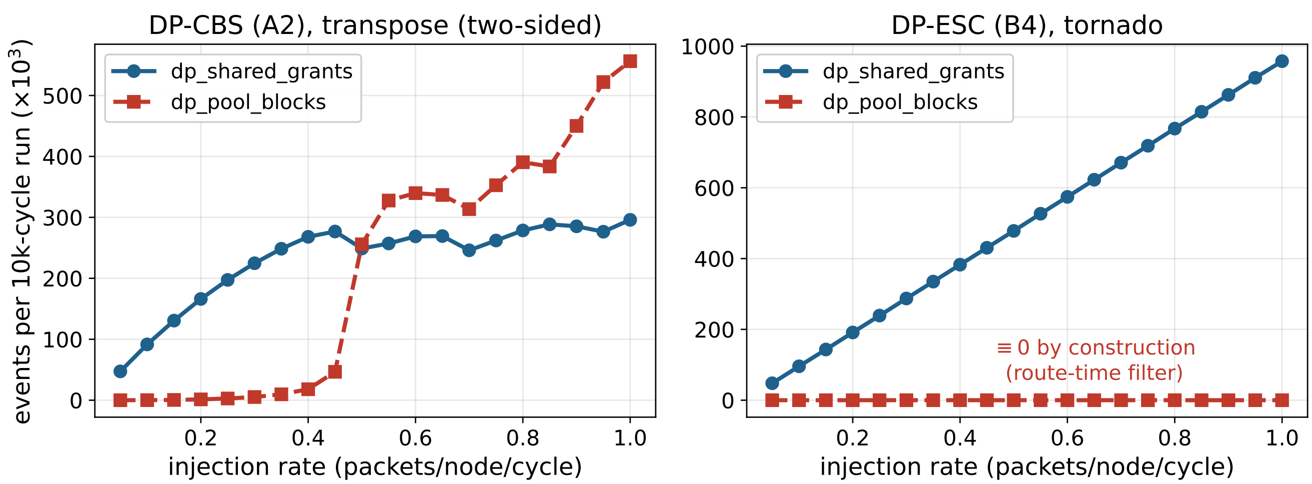

Figure 6: Pool activity vs. injection rate. Left: DP-CBS (A2) on two-sided transpose — the pool is heavily exercised (`dp_shared_grants`) and cap denials (`dp_pool_blocks`) surge once the two directions start competing past saturation, confirming the counter works. On one-sided patterns the same counter is exactly zero: a denial requires the opposite side to hold budget. Right: DP-ESC (B4) on tornado — `dp_pool_blocks` is identically zero *by construction*, since pool pressure is absorbed by the route-time filter.

6 Analysis

6.1 Why pooling wins

Three layers explain the gains:

1. **Statistical multiplexing of buffer capacity.** Pooling converts private idle capacity into shared usable capacity; its value is exactly the demand imbalance between the two directions. This predicts the observed pattern dependence: adversarial one-sided patterns gain 20–31%, symmetric uniform random gains ≈0.

2. **The concrete bottleneck is hot-inport lane count × credit round trip.** On this substrate a VC holds one packet and turns around in ∼5 cycles, so each usable lane on the hot inport is worth ≈0.2 packets/cycle of saturation throughput. DP raises the number of lanes the hot direction may *occupy* (dedicated + whole

cap instead of dedicated + half the storage), a latency-hiding effect analogous to adding MSHRs to a cache. The family-B peak progression matches this model quantitatively: B1 saturates near 0.195 with one usable adaptive lane per inport, B2/B3 near 0.40 with two, and B5 at the 0.5 ceiling with three.

3. **Separation of safety and performance resources.** What makes the first two layers *safe to exploit* is that the pooled tranche carries no deadlock-freedom obligation: the substrate (CBS sub-ring or escape VCs) owns the invariants, so admission can fail arbitrarily. This separation is the design's transferable idea.

6.2 Costs and regime boundaries

Cap contention (reverse blocking). Pooling's definitional cost: under two-sided load, packets of one direction can exhaust the budget and deny the other direction admission to its own physically free pooled VCs (-32.6% worst case at equal VC count). Three mitigations bound it: the unpooled substrate guarantees a progress floor for the denied side, DP-ESC reroutes around full pools at route-computation time, and occupancy leaks back within a credit round trip ($\sim$5 cycles). Per-side fairness under sustained asymmetry (aging, per-side floors beyond the substrate) is left as future work.

Buffer-rich regimes. At C=8 every mechanism drowns in buffers: DP-CBS turns slightly negative and family B saturates the harness ceiling. DP targets scarce-buffer designs.

State staleness. The simulated registry is a zero-latency oracle. A real sideband adds 1–2 cycles of skew; since the accounting window already includes the credit round trip and is conservative, safety is unaffected, and a token-conserved realization brackets the modeled performance from below. The headline one-sided regimes are insensitive (tokens park at the hot side); the marginal cells (+1–3%) may erode toward zero.

Substrate specificity. With one-flit control VCs, every extra usable hot-side slot is worth a full 1/RTT of throughput; deeper VCs would shrink the magnitudes (the direction of the effect stands). Five of the fifteen cells in Table 3 sit at the injection ceiling on both sides and say nothing about relative headroom, and the "$\geq$" cells are truncated by the same ceiling on the DP side; single-seed cells at $\pm$1–3% are reported as zero-cost bounds, not gains.

7 Conclusion

Dimension Pool shows that the static per-direction split of router buffers is a real inefficiency on tori and that it can be removed with a counter-only occupancy budget — provided the deadlock-freedom invariant lives in a resource the pool can never touch. Composed with CBS or with escape VCs on a $4\times4\times4$ Garnet Torus3D, DP delivers +21–31% (DP-ESC, C=6, all patterns) and +20.9% (DP-CBS, C=6, xopposite) peak stable throughput at equal usable storage, is neutral on symmetric traffic, deadlock-free across 1006 runs, and comes with explicitly mapped costs: cap contention under two-sided load, no benefit in buffer-rich regimes, and a wiring sideband as the honest hardware price.

References

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